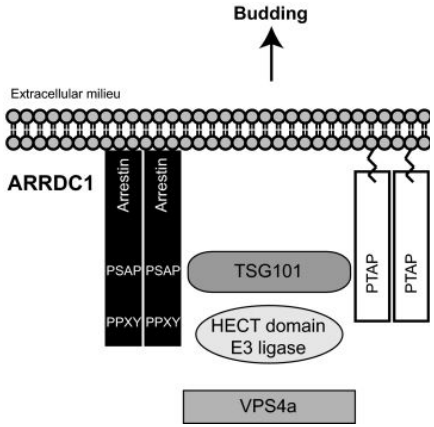
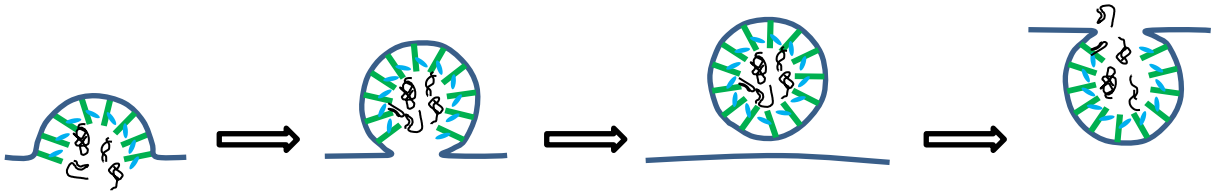
ARMMS for Dual shRNA Delivery against SARS-CoV-2 and Enterovirus

Diane Ting and Jerome Wu

What are ARMMs?

Arrestin Domain Containing Protein 1 [ARRDC1]-Mediated Microvesicles





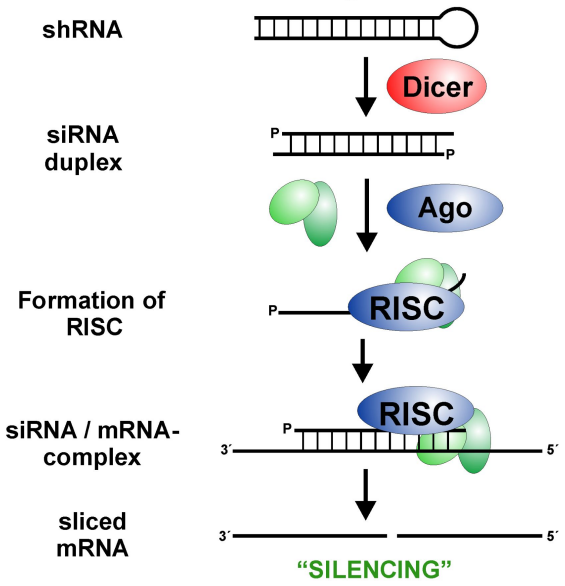
ARRDC1/Tsg101 initiates cell surface budding

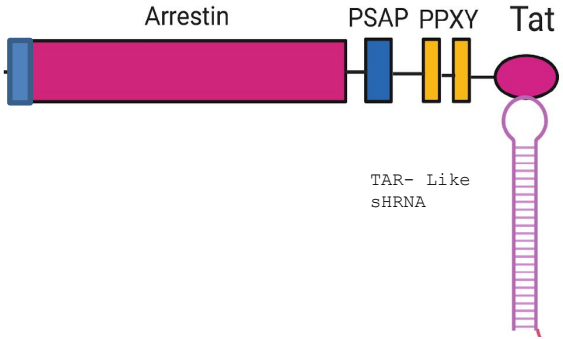
Tsg101/ESCRT promotes neck/stem formation

ESCRT/Vps4 initiates scission and EVBPM release

Direct fusion with plasma membrane of target cell

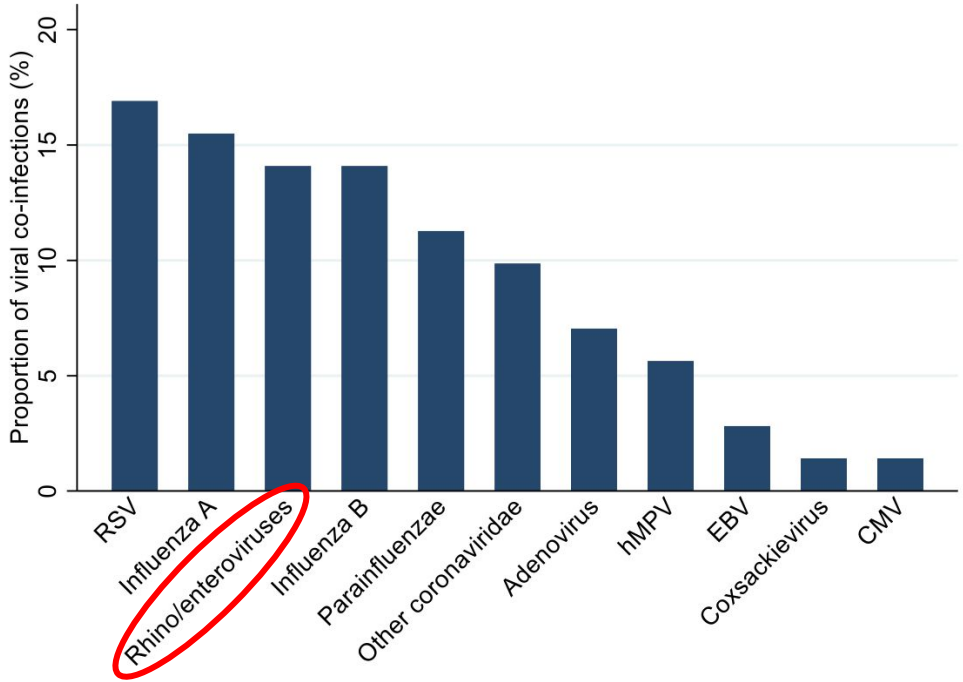
shRNA


shRNA
Dicer
siRNA duplex
P
P
Ago
Formation of RISC
P
RISC
RISC
siRNA / mRNA-complex
P
3´
5´
sliced mRNA
3´
5´
"SILENCING"


Arrestin
PSAP
PPXY
Tat
TAR- Like sHRNA

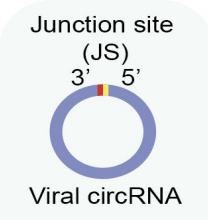
Co-Infection

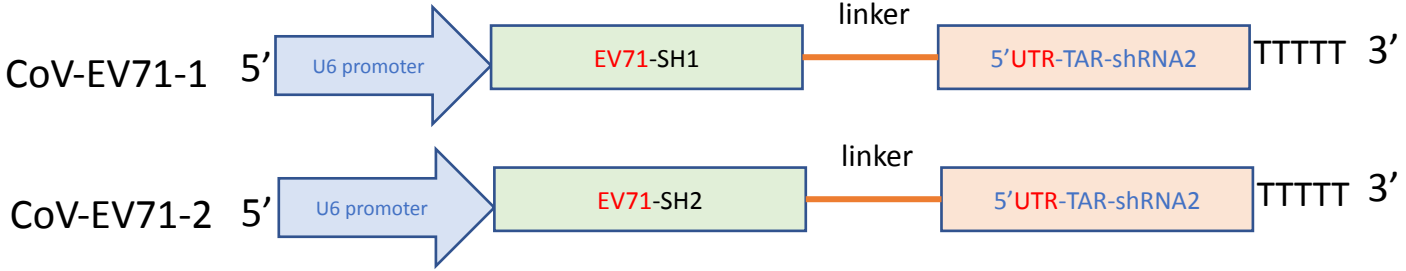
Covid Viral Coinfection affects 4-40% of patients



Lim et al (2020)

Enterovirus





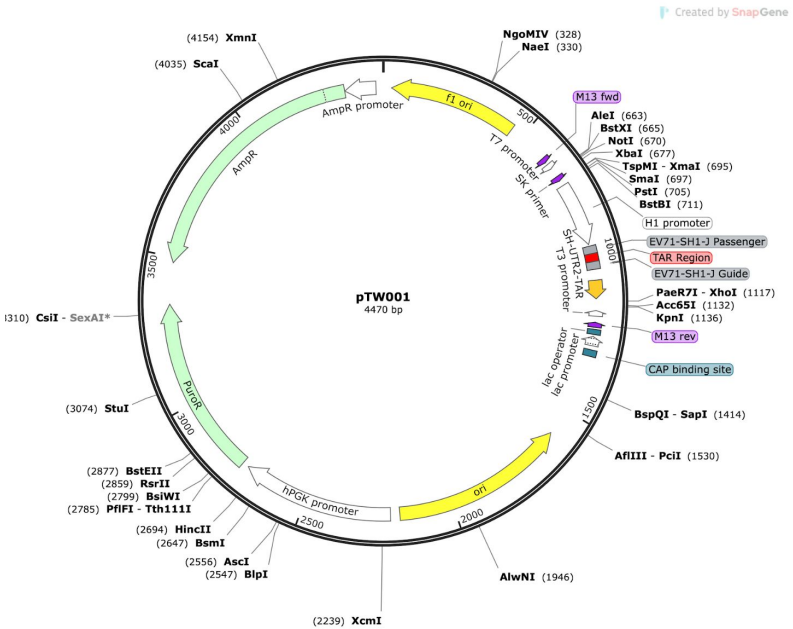
>EV71_3226/3375 (+) stranded

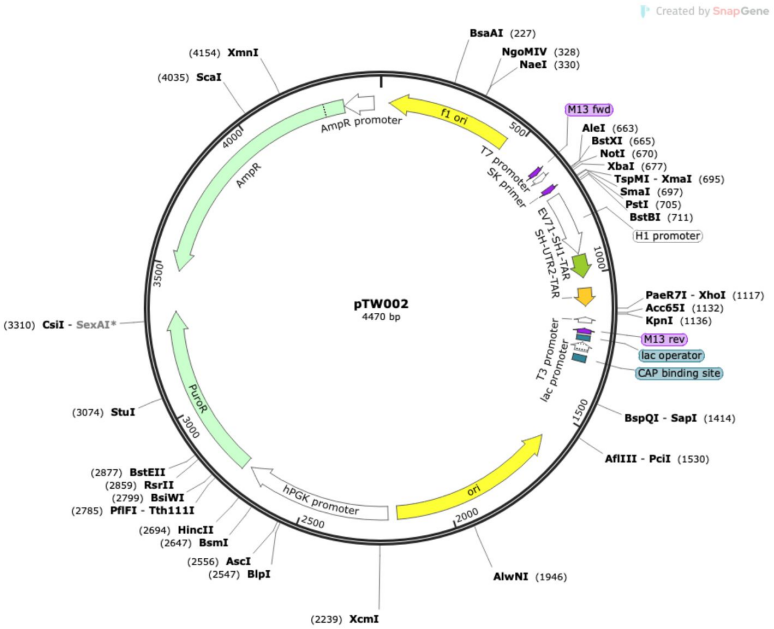
EV71-SH1

CTGGTGCCAGTCGCACAGCAATCACCACCCTCGGGAAATTTGGACAGCAGTCCGGAGCTATCTACGTGGGCAACT
T|ACCTCGCCCAATGCGTAACCAGAACTATCTATTCAAAGCCAACCCAAATTATGCTGGTAATTCTATTAAACCAA

EV71-SH2

3375|3226
Junction site

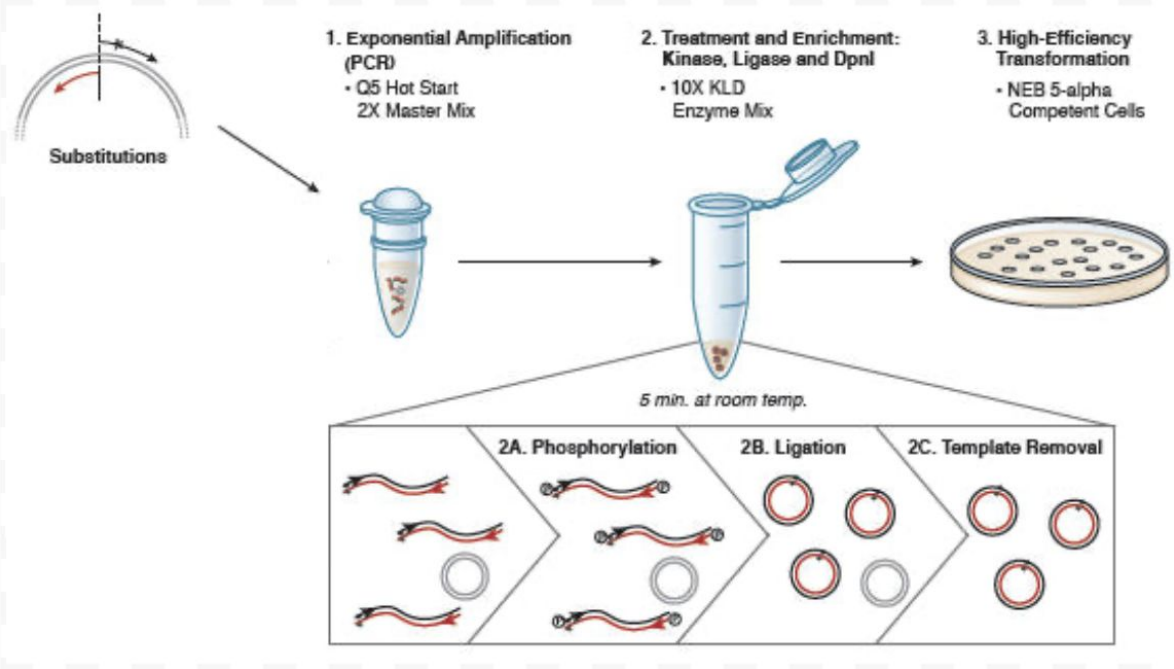




Our Constructs

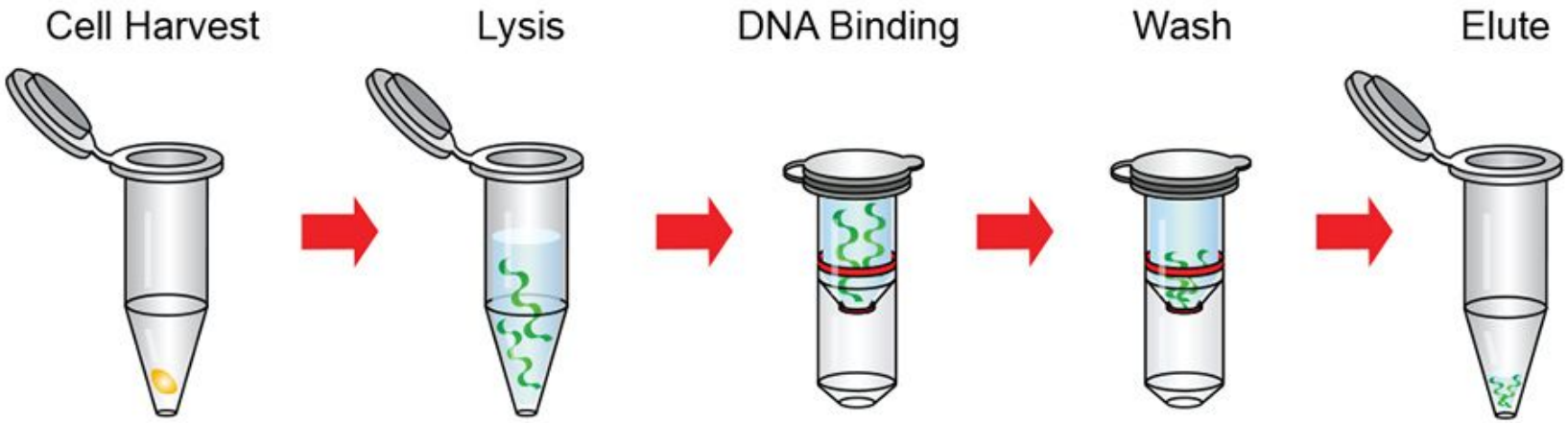
Methodology

Mutagenesis



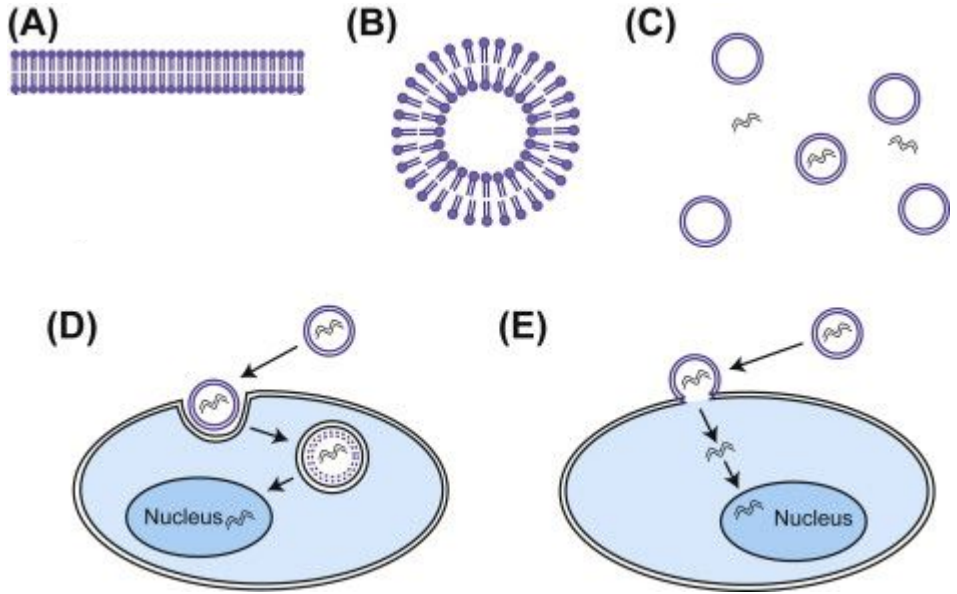
- PCR
- Create KLD
- Transformation

MiniPrep



https://www.genedirex.com/product/plasmid-miniprep-kit/

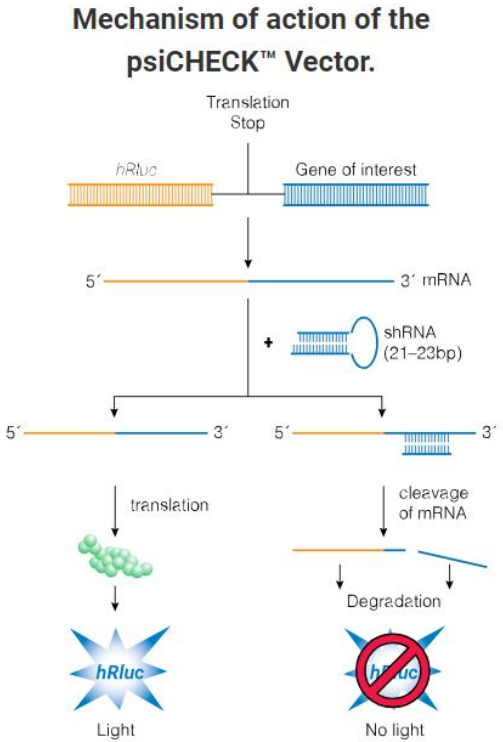
Transfection

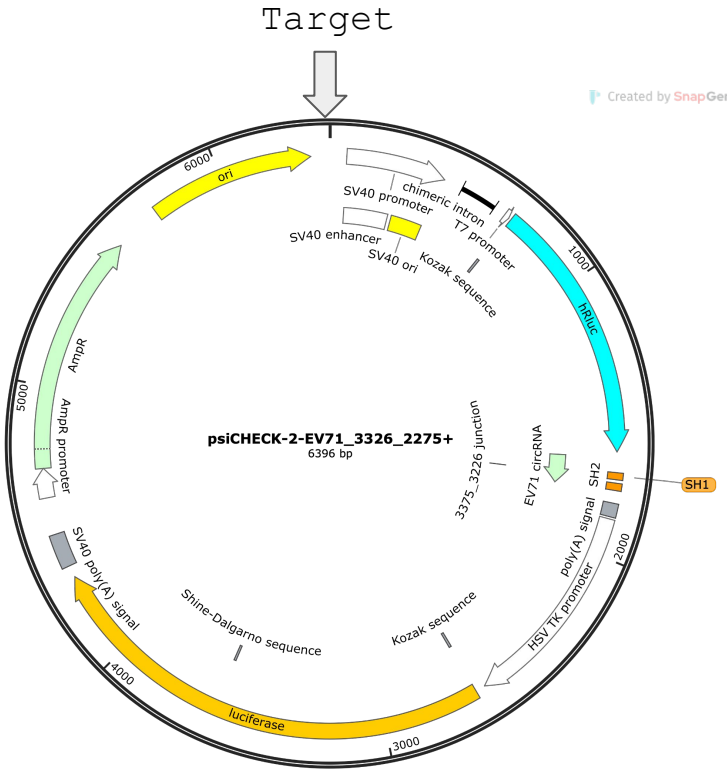


- Done via Tissue Culture with HEK293 Cells and Clone 5
- Insert constructs into cells

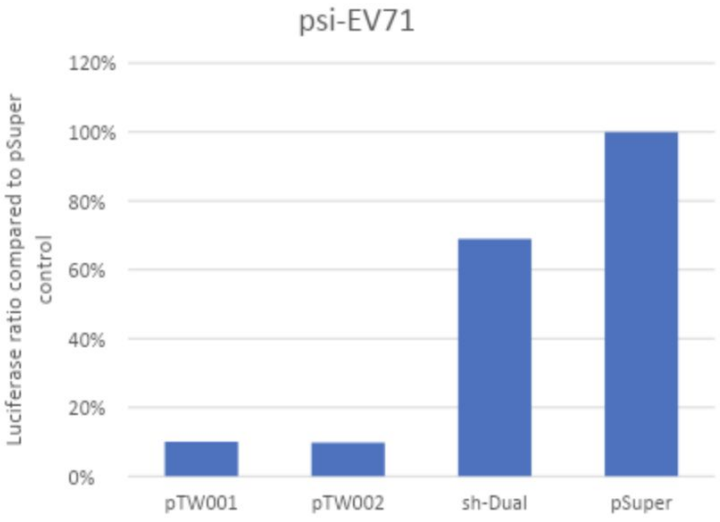
https://www.sciencedirect.com/book/9780128005118/guide-to-research-techniques-in-neuroscience

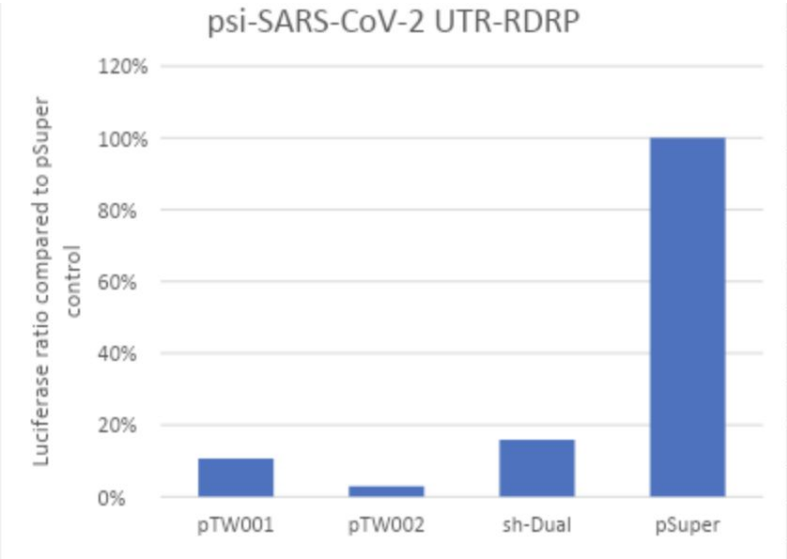
Luciferase Mechanism





Luciferase Results

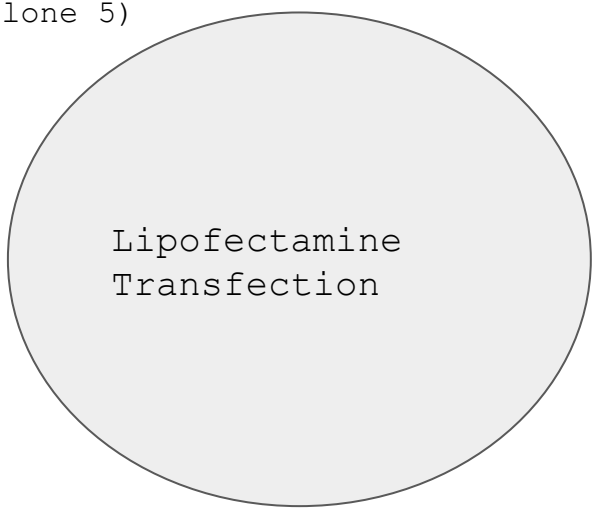




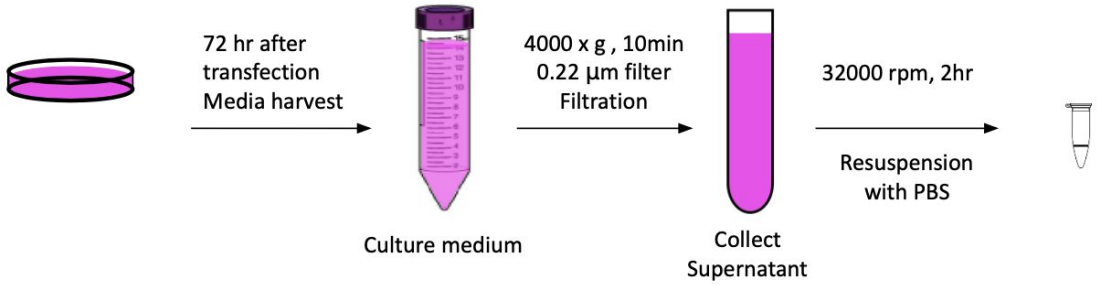
ARMMS Production and Collection

Production

NoDicer ARRDC1-expressing Cell (Clone 5)



Collection



What's Next?

- Verify knockdown efficiency of ARMMS
- Pre-Clinical Trials: Demonstrate viral inhibitory effect of ARMMS
- If all goes well, this may go market one day =)

Acknowledgements

- Cohen Lab: Dr. Yanan Feng, Dr. Weijing Xu, Xiaohong Tang, Professor Cohen and Ms. Roberta Peterson

- Dr. Peter Sarnow

- Our lovely GRIPS admin Mrs. Dawn Billman and Mr. Robert Monroy along with our many guest speakers

References

1) Lansbury, L., Lim, B., Baskaran, V., & Lim, W. S. (2020). Co-infections in people with COVID-19: A systematic review and meta-analysis. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3594598

1) Nabhan, J. F., Hu, R., Oh, R. S., Cohen, S. N., & Lu, Q. (2012). Formation and release of arrestin domain-containing protein 1-mediated microvesicles (ARMMs) at plasma membrane by recruitment of TSG101 protein. *Proceedings of the National Academy of Sciences*, *109*(11), 4146-4151. https://doi.org/10.1073/pnas.1200448109

2) Wang, Q., Yu, J., Kadungure, T., Beyene, J., Zhang, H., & Lu, Q. (2018). ARMMs as a versatile platform for intracellular delivery of macromolecules. *Nature Communications*, *9*(1). https://doi.org/10.1038/s41467-018-03390-x